

Panayotis - Philip Papavassilopoulos

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EDUCATION

University of Southern California (USC)

May 2029 (Expected)

PhD, Viterbi School of Engineering

- Area of study: Industrial & Systems Engineering (ISE) / Operations Research, Machine Learning & Artificial Intelligence.

Research focus: Mathematical Optimization with applications in Machine Learning, Transportation, Logistics, Economics

University of Illinois Urbana-Champaign (UIUC)

May 2026

Master of Science, Grainger College of Engineering

Grade: 3.92/4.0

- Area of study: Industrial & Systems Engineering (ISE) / Operations Research.

Related Coursework: Linear Programming, Integer Programming, Nonlinear Programming, Optimization Under Uncertainty, Real Analysis, Game Theory, Network Optimization, Statistical Learning Theory, Probability Theory and Statistics, Stochastic Processes

National Technical University of Athens (NTUA)

June 2023

Bachelor of Science & Master of Science (5-year combined degree)

Grade: 7.62/10

- Area of study: Electrical and Computer Engineering.

Related Coursework: Multiple Criteria Decision Analysis, Analysis of Data, Forecasting Techniques, Algorithms and Complexity, Operating Systems, Neural Networks and Machine Learning, Data Analysis/ Database Systems, Web/ Mobile App Development

GRE Quantitative score: 168/170

PUBLICATIONS

- Panayotis P. Papavassilopoulos, Grani A. Hanasusanto, and Yijie Wang. A Distributionally Robust Optimization Approach to Quick Response Models under Demand Uncertainty.
(Submitted for journal publication)

Description: Formulated and solved a distributionally robust inventory model accounting for ambiguity in demand distributions. Derived tractable optimization reformulations, implemented large-scale computational experiments using CVXPY and demonstrated significant reductions in the waste-to-consumption ratios across uncertainty scenarios.

- Y.C.Yoon, P.Papavassilopoulos, G.A. Hanasusanto, and Yijie Wang. Data-Driven Distributionally Robust Contextual Optimization with Gaussian Mixture Models.
(Accepted and presented at the XVII International Conference on Stochastic Programming (ICSP), Paris, France, July 31, 2025)

Description: Developed a contextual stochastic optimization framework using Gaussian Mixture Models and distributionally robust methods, achieving superior sample complexity and out-of-sample performance across high-dimensional decision problems.

- Bachelor Thesis: *Assessment of Sustainable Development Goals with Integrated Assessment Models in the Energy field.* (in Greek).

Description: Developed a method to estimate minimum objective-function ranges in multi-objective optimization problems, improving both robustness and computational efficiency. Also applied this method to determine optimal investment policies for renewable energy sources.

EXPERIENCE

University of Illinois Urbana-Champaign

Urbana-Champaign, Illinois, USA

Graduate Research Assistant

- Developed tractable two-stage stochastic and robust optimization models for data-driven decisions in manufacturing, supply chains, and transportation logistics.
- Applied a two-stage stochastic optimization model to sustainable fashion production, demonstrating reductions in production costs and water consumption while satisfying environmental regulations.
- Developed a robust truck-drone optimization model for post-disaster relief distribution, incorporating decision-dependent demand for food and medical supplies.

Teaching Assistant

September 2024 - May 2026

Courses (2): "Computing for Industrial and Systems Engineering", "Business Side of Engineering"

- Contributed to the creation and refinement of Midterm and Final exam problems.
- Conducted office hours to clarify and expand upon classroom concepts for students, graded exams and homework assignments.

Decision and Support Systems Laboratory (NTUA)

Athens, Greece

Undergraduate Researcher

September 2022 - August 2023

- Analyzed policies by assessing Sustainable Development Goals with Integrated Assessment Models, used Multiple Criteria Optimization methods and Algorithms.

Startup Business SoFiLy

Athens, Greece

Co-founder

April 2021 - June 2023

- Developed a fitness-related social network app using computer vision and body tracking technology in motion games.
- Conducted research to formulate commercial strategies and facilitated cross-functional collaboration between the business and technical teams.

Stochasis Management Consultancy S.A.

Athens, Greece

Intern

January 2022 - December 2022

- Conducted data collection and analysis for Greek Economy's Primary and Secondary sectors, contributing insights to studies and reports.

Control Systems and Robotics Laboratory (NTUA)

Athens, Greece

Lab Assistant

March 2021 - June 2021

- Engaged in lab experiments and contributed to the development of educational materials used in student exercises.

Volunteer Private Tutor

Athens, Greece | September 2018 - May 2021

- Provided one-on-one tutoring students in high-school level mathematics, physics, and informatics.
- Adapted the lessons to individual learning styles, communicated complex concepts in a clear and understandable manner.

PROJECTS, HONORS & AWARDS

Fellowships

August 2023 - June 2024

- William A. Chittenden II Fellowship
Full scholarship and tuition fee waiver for the 2023-2024 academic year *awarded by the University of Illinois Urbana-Champaign for academic excellence.*
- Jerry S. Dobrovolny Fellowship
Awarded by the University of Illinois Urbana-Champaign for academic excellence.

'CodeBiz' National student competition (Team 1st place)

Athens, Greece, April 2021

- Student competition which combines business skills with innovation and programming.
- Developed a social networking app utilizing motion tracking and machine learning, promoting physical activity and socialization where users could record, mimic, and engage in motion-based games.
- Post-competition, adapted the app for use in injury restoration programs, showcasing its practical application in healthcare.

PERSONAL SKILLS

Computer Skills:

- Programming Languages: C, C++, Python, Java, JavaFX, SQL, Swift, XML, MatLab, Node.js.

Software Skills: Unix, Apache Spark, Hadoop, Gurobi, CVXPY,

Spoken Languages: Greek (Native), English (Fluent), French (Intermediate)

SOCIETIES & AFFILIATIONS

Hellenic Students Association of UIUC (August 2023 - present).

Urbana-Champaign, Illinois, USA

Vice-president

August 2023 - Present

- Contributed to organizing and participating in events designed to connect Greek students and community members in Urbana-Champaign.
- Promoted Greek customs and traditions throughout the year, fostering a shared cultural identity and preserving Greek heritage.

Model United Nations (MUN) NTUA Team

Thessaloniki, Greece

Delegate

May 2017

- Shaped policies as a United Nations delegate during MUN competitions to address real-world challenges.

2nd Scout System of Papagos

Athens, Greece

Boy Scout

September 2005 - June 2014

- Organizing and executing interactive activities, leading small group into excursions and camping.

HOBBIES & SPORTS

Piano, Long Distance Running, Camping, Hiking, Sailing, Cooking